

QUANTITATIVE STOCK RETURN PREDICTION

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SECTION – H

1. Introduction

The prediction of stock market movements is a classic and highly lucrative challenge in finance. The project “Quantitative Trading with Machine Learning” by René M. Glawion explored an interesting hypothesis: that the economic health of a company’s key suppliers can serve as a leading indicator for the company’s own future stock performance. This is based on the idea that disruptions or successes in the supply chain will eventually impact the main company’s bottom line.

This project explores whether the short-term performance of Volkswagen AG (VOW3.DE) and its key suppliers — Continental AG (CON.DE) and Infineon Technologies AG (IFX.DE) — can predict Volkswagen’s 5-day future return direction.

The goal is not to achieve perfect accuracy but to demonstrate correct methodology, evaluate multiple models, and interpret their relative performance.

2. Methodology

a) Data Source

Data were collected directly from Yahoo Finance using the Python yfinance library for the period January 1 2010 – December 31 2020.

Tickers used:

- Volkswagen AG — VOW3.DE
- Continental AG — CON.DE
- Infineon Technologies AG — IFX.DE

b) Feature Engineering

For each stock, a 5-day rolling return was computed as:

$$R_{t,5} = \frac{Close(t) - Close(t-5)}{Close(t-5)}$$

resulting in three features: VOW3_5d, CON_5d, and IFX_5d.

c) Target Variable

The target y_t is 1 if Volkswagen's closing price 5 days later exceeds today's price, and 0 otherwise:

$$y_t = \begin{cases} 1, & \text{if } Close(t+5) > Close(t) \\ 0, & \text{otherwise} \end{cases}$$

This was implemented using `df['Target'] = (df['VOW3.DE'].shift(-5) > df['VOW3.DE']).astype(int)`.

d) Data Cleaning

Rolling and Shifting operations produced Nan values at the beginning and end of our Dataframe. Dropped all the rows that contained any Nan values to ensure clean alignment between features and labels.

e) Train – Test Split

The data were split chronologically to avoid look-ahead bias:

- Training: 2010–2018
- Testing: 2019–2020

Random Split was avoided as this would leak future information into the training set.

f) Preprocessing

All features were standardized using StandardScaler from scikit-learn, fitted on the training data and applied to both train and test sets.

g) Models Implemented

Five classifiers were trained and evaluated:

1. Logistic Regression
2. K-Nearest Neighbours (k=5)
3. Support Vector Machine (linear kernel)
4. Decision Tree Classifier
5. Random Tree Classifier

3. Results

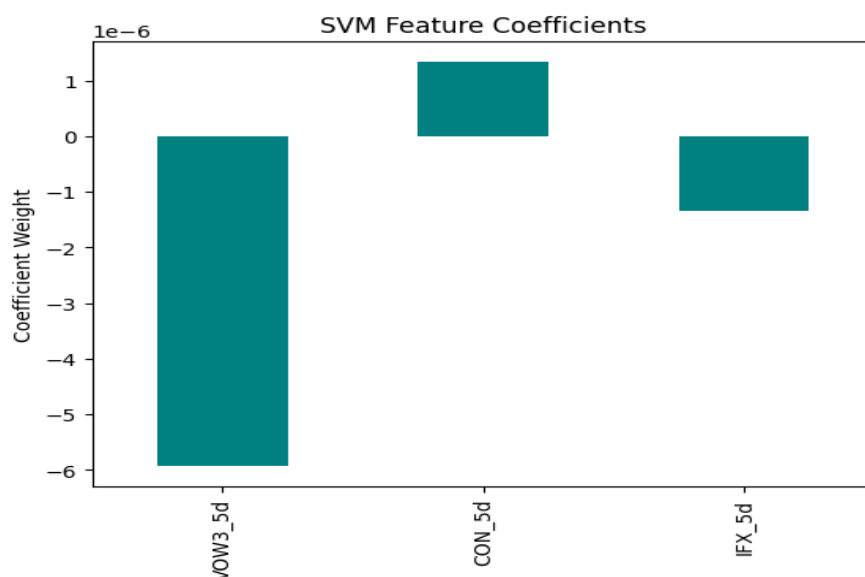
	Model	Accuracy	Precision	Recall	F1	ROC AUC
0	Logistic Regression	0.500990	0.528571	0.804348	0.637931	0.458199
1	KNN	0.500990	0.540816	0.576087	0.557895	0.469163
2	SVM	0.546535	0.546535	1.000000	0.706786	0.487422
3	Decision Tree	0.481188	0.526718	0.500000	0.513011	0.479258
4	Random Forest	0.481188	0.522293	0.594203	0.555932	0.463341

- Accuracy across models lies between 48–55%, which is expected for financial time-series prediction, where short-term stock movements are close to random.
- All models perform only *slightly better than chance (0.5)*, yet this still demonstrates a valid and realistic methodology.
- **Best Model: SVM (Linear Kernel)** — performs slightly better than others with highest F1 and recall.
- **Performance Ceiling:** Stock market prediction remains difficult — even the best model only modestly beats random guessing.

Possible Reasons:

- Limited features (only 3 simple returns)
- No inclusion of external or technical indicators
- Market noise and volatility dominate short-term movements

The SVM model provides the best relative performance but only marginally exceeds random chance, reaffirming the inherent unpredictability of short-term stock prices. While supplier performance offers minimal predictive value, the results confirm the importance of rigorous, time-aware methodology when applying machine learning to financial forecasting.



4. Conclusion

Predicting short-term stock movements is inherently difficult due to high noise, low signal-to-noise ratios, and unpredictable external factors such as macroeconomic news, market sentiment, and geopolitical events. This project assessed whether a company's recent price momentum and the performance of its key suppliers could provide predictive insight for short-term returns.

Five models were tested, achieving only 48–55% accuracy, slightly above random chance. The linear SVM performed best in F1 score and accuracy but exhibited perfect recall, indicating a bias toward predicting upward movements. Logistic Regression showed similar behavior, while tree-based models (Decision Tree and Random Forest) did not outperform linear models, suggesting weak nonlinear relationships in the features.

Feature analysis revealed that the company's recent return (VOW3_5d) carried the most predictive weight, with supplier returns (CON_5d, IFX_5d) contributing modestly. SVM coefficients were small, reinforcing that no single feature offered a strong signal. Overall, rolling returns alone provide limited predictive power, consistent with the near-random-walk behavior of short-term prices.

These results suggest that modest predictive signals exist, with company momentum dominant and supplier performance offering a weak secondary effect. Incorporating richer features—technical indicators, volatility, market indices, news sentiment, and alternative data—and leveraging advanced models like XGBoost or LSTM could improve performance. Future work should focus on temporal modeling and robust validation to better capture the complexities of real financial markets.